

LSGI3212

Intelligent Transportation Systems

Simulating Hong Kong Transportation Activities with Customized MATSim Scenarios

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1. Abstract

Multi-Agent Transport Simulation (MATSim) software is a framework for simulations of different settings of the transportation system available to the public. With the modification and creation of the road network, the performance of the proposed transportation system can be simulated and evaluated. Therefore, it has become a popular simulation software to explore travel behavior and congestion patterns. It is also frequently used in studies related to transportation systems, such as the research on analyzing the energy consumption of transportation through modelling with MATSim (Novosel et al., 2015).

Nowadays, long queues of slow-moving traffic have increasingly become a more common scene in Hong Kong. This report explores the use of MATSim (Multi-Agent Transport Simulation), to better represent the transportation activities in Hong Kong by developing our customized scenario. We start with the base case – the “equil” scenario provided by MATSim, which has a simple network and agent daily plans. We have made modifications to incorporate more realistic characteristics based on a 2011 travel survey in Hong Kong. The results of our customized scenario compared to the “equil” base case can help us understand how the modifications influence the simulation outcomes.

2. Background

At MATSim, agent-based transport models describe a synthetic population of agents which are represented by the travelers in terms of their demographic information and travel demand. Each agent has at least one plan that indicates the sequence of commuting with each pair of activities linked by a link. Also, the capacity, length and number of the links enable the simulator to acquire the knowledge of physical space for vehicles to access. Specifically, the aggregation and execution of activities of these agents can model 24-hour transportation scenarios for a population of agents. The framework provides a base "equil" scenario, which can be customized to represent specific networks and populations, making it a widely used tool for urban transport simulations.

Hong Kong is a densely populated urban area with unique transport challenges. In this project, we modify the "equil" scenario to simulate Hong Kong's transport network and real-world travel behavior based on the 2011 travel survey data. Thus, we can conduct detailed analyses of urban mobility patterns and infrastructure performance.

3. Introduction to MATSim

3.1 Framework

According to Horni et al. (2016), the MATSim framework can be divided into three stages: input of the initial transportation plan of agents, simulation process, and output section. The simulation process can be further divided into execution, scoring, and replanning stages.

The unit of analysis in MATSim is a single-day activity-based model. The initial inputs include the daily activity schedule, facilities and transportation networks of each agent. These inputs provide the necessary details for simulating agent activities and travel between locations. Then, in the execution module, the daily plan of each agent is executed by simulation of activities performance and travel between activity locations. For example, one agent may drive from home to work at 8 AM and then to the market at 6 PM, while another may cycle to school in the morning and return home in the afternoon.

To evaluate the performance of each agent, a utility function of activity and travel is used. By adjusting plan elements, such as departure time and route, the daily plan for each agent is optimized by a configurable number of iterations. Various output files including link statistics, score statistics, traffic flow data are generated for analysis. Then, each agent will be scored based on the performance of his transportation plan executed.

After the scoring process, the plans for each agent will be adjusted. First, for the agents with plans exceeding the maximum limit, the plans will be evaluated and those with better performance (higher score) are selected. For designed agents, users can configure numerous parameters to specify the scoring function for the next mobsim loop. By analyzing the scores of the daily plans, the replanning module adjust them in terms of elements such as transport mode, departure time, transportation route, etc. As a result, the plan with highest score will be selected, as it is the most preferred plan.

The simulation system runs again to find out the best plans for the agents, meaning the plans with the highest scores, as the output of the programme. The traffic condition, such as the traffic flow during rush hours is extracted in this project, and the optimized daily plan for the travelling agents will be produced.

3.2 Mechanisms

The performance of the simulated daily plans for each agent is evaluated in this module with the use of a utility function (Horni et al., 2016). After simulating the agents travelling from different locations at different times, the utility function is applied to assess the travel plan for each agent. The basic form of the function is shown below, with the score of a daily plan consisting of the score of the activity utilities and the travel utilities:

$$S_{plan} = \sum_{q=0}^{N-1} S_{act,q} + \sum_{q=0}^{N-1} S_{trav,mode(q)}$$

Figure 01
(Horni et al., 2016)

The utility of all the activities (first part of the function), includes scores of various elements, which are the duration, waiting duration, late arrival, early departure, and short duration. The elements carry positive and negative effects to the activity utility, which will be elaborated in the following paragraphs.

$$S_{act,q} = S_{dur,q} + S_{wait,q} + S_{late.ar,q} + S_{early.dp,q} + S_{short.dur,q}$$

Figure 02

(Horni et al., 2016)

The first element, duration, represents the duration of the activity. In MATSim, each activity, such as work, leisure, shopping, etc. carries a different score and the duration of performing the activity is taken in account of the utility score. For example, the system can grade ‘working’ with a basic score of 10 and ‘leisure’ with a basic score of 5. The next element, waiting duration, means the score of the agent waiting to perform the activity. As it is the time wasted for the agents to wait to begin the execution of the activity, this element carries a negative effect. For example, an agent departs from home to shop in the market, which opens at 11 am. The wait element is taken into account when the agent arrives at the market, like at 10:50 am, so he has to wait outside the market until 11 am to start executing shopping activity. Third, late arrival demotes the late start of the activity, so it also carries a negative scoring effect. It stands for the score deduction like when the football game starts at 6 pm, but the agent arrives at the stadium at 6:30 pm. For early departure, it stands for the time when the agent finishes the activity earlier than the planned time. For example, to avoid the traffic congestion during rush hour, the agent leaves his office an hour earlier than the finishing time of his work, then the utility score will be deducted. The last element is the short duration, which specifies the negative utility factor of having an activity that requires time shorter than the specified shortest duration for each activity.

$$S_{trav,q} = C_{mode(q)} + \beta_{trav,mode(q)} \cdot t_{trav,q} + \beta_m \cdot \Delta m_q + (\beta_{d,mode(q)} + \beta_m \cdot \gamma_{d,mode(q)}) \cdot d_{trav,q} + \beta_{transfer} \cdot x_{transfer,q}$$

Figure 03

(Horni et al., 2016)

The above equation shows the travel utility function. It is used to calculate the utility score of the travels, taking into account factors like travel time spent, transport mode, transport fees with a penalty for over-budget, travelling distance as affected by the routes, etc. It evaluates how good or bad the daily plan is in terms of transportation. For example, travelling by car deducts less points as the travelling time is usually shorter, but there may be a penalty as the cost is higher than riding a bike.

4. Introduction of the base MATSim scenario

The MATSim scenario adopted in this project is constructed upon an existing base MATSim scenario. The “equil” scenario is selected for the purpose of this analysis. All the four essential files of the base scenario are used, including file config, facilities, network, and plans.

The "equil" scenario network comprises 15 nodes and 23 facilities. It employs the plan100.xml file to configure 100 agents, all of whom start and end at the same location and share identical travel purposes. In Figure 4, the dark grey lines depict the original road links in the adopted base scenario, while the facilities in the network are represented by purple triangles.

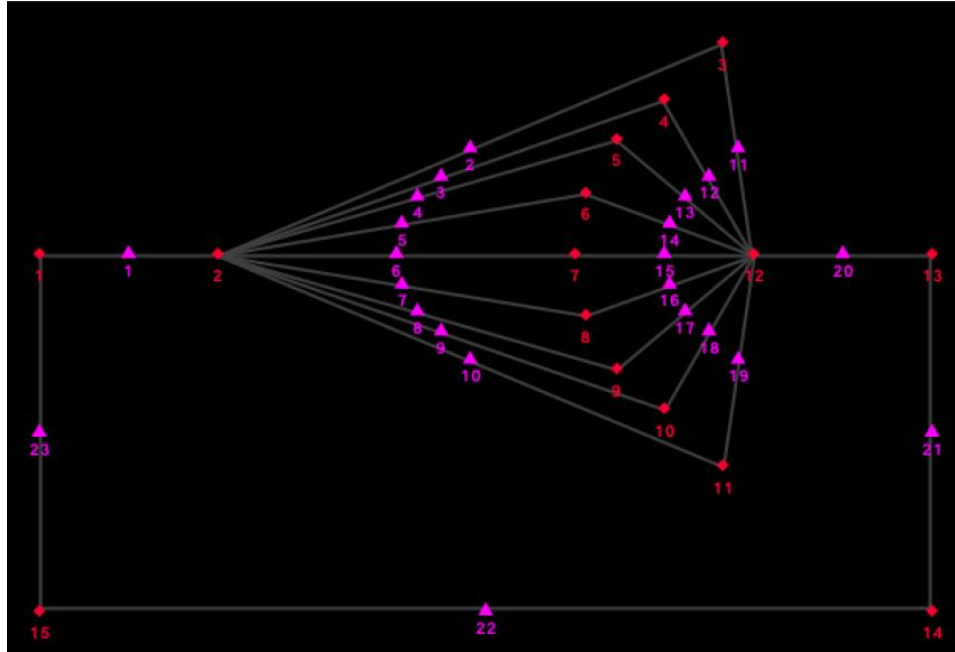


Figure 04. Base MATSim Scenario

The "equil" scenario is selected as it serves as an ideal starting point for our customized Hong Kong transportation simulation due to its fundamental characteristics and adaptability. It provides a controlled environment for implementing our customizations. This baseline simplicity allows us to gradually introduce complexity by adding diverse travel purposes and varying multiple parameters, while maintaining the ability to trace the impact of each modification.

Additionally, this scenario offers clear visibility of cause-and-effect relationships when modifications are implemented with its basic structure. Its simplified network configuration allows us to understand how each modification impacts the simulation results. This transparency is particularly valuable when testing different parameters and validating the simulation behavior.

5. Customized MATSim scenarios

To better represent the transportation activities in Hong Kong, we make several modifications to the base scenario:

- We use a sample of 500 agents, which is the maximum number of agents that can be visualized using the free license of the Via visualization tool.
- Based on the 2011 Hong Kong travel characteristics survey, people travel not only because of work, but as well as for school or other reasons. We assume that the people live close to the links, and their destinations are close to the facilities to simulate the dense urban environment of Hong Kong. According to the survey, the mean walking time from trip origin to mechanized transport mode was 4 minutes. We generate the agent daily plans (`path500.xml`) using a Python script that randomizes the departure times, durations, and locations of work, school, and other activities based on the survey (Travel Department, 2011).
- We characterized our data into multiple parameters in `consts.py` that represents our agent population model. We use different weights to generate the travel purpose, and two sets of normal curves to generate the travel start and end timestamps. We generate 500 agents to maximize the number of agents visualized in Via. We also implemented a graph plotter in Python which can help us to visualize the agent population, travel purposes, and trip-making time. We make fine adjustments to the parameters to make the graph overlap the graph from the Transport Department as much as possible.

```
consts.py x
1  from datetime import time, timedelta
2
3  MAX_DIST_FROM_FACILITY: int = 150
4  MAX_DIST_FROM_LINK: int = 150
5
6  TRAVEL_INTENTS = ['w', 's', 'o']
7  TRAVEL_WEIGHTS = [0.5, 0.1, 0.4]
8
9  WORK_DEP_TIME_MEDIAN = time(8, 0)
10 WORK_DEP_TIME_SD = timedelta(hours=1)
11 WORK_END_TIME_MEDIAN = time(17, 0)
12 WORK_END_TIME_SD = timedelta(hours=1.5)
13 SCHOOL_DEP_TIME_MEDIAN = time(7, 30)
14 SCHOOL_DEP_TIME_SD = timedelta(hours=0.5)
15 SCHOOL_END_TIME_MEDIAN = time(15, 0)
16 SCHOOL_END_TIME_SD = timedelta(hours=0.5)
17 OTHER_DEP_OR_END_TIME_MEDIAN = time(15, 0)
18 OTHER_DEP_OR_END_TIME_SD = timedelta(hours=4)
19
20 FILE_NAME = "plans500.xml"
21 # Note: maximum agents in Via is 500
22 AGENT_COUNT = 500
23
```

Figure 05

5.1 Modification of input data

In order to achieve the objectives of this project, the base MATSim scenario is modified in a couple ways to provide the desired outcome. The modifications to the base MATSim scenario will be described and explained below.

5.1.1 Modification of network data and reasons

Figures 6, 7 and 8 show the content of network.xml in scenario 1, scenario 2 and scenario 3 respectively.

[illegible]

Figure 06.

[illegible]

Figures 07.

```
<!--links capexid="01:00:00"-->
<link id="101" from="1" to="2" length="1500.00" capacity="80" freespeed="14.88" permlanes="1" />
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<link id="103" from="2" to="4" length="1500.00" capacity="40" freespeed="14.88" permlanes="1" />
<link id="104" from="2" to="5" length="1500.00" capacity="40" freespeed="14.88" permlanes="1" />
<link id="105" from="2" to="6" length="1500.00" capacity="40" freespeed="14.88" permlanes="1" />
<link id="106" from="2" to="7" length="1500.00" capacity="40" freespeed="14.88" permlanes="1" />
<link id="107" from="2" to="8" length="1500.00" capacity="40" freespeed="14.88" permlanes="1" />
<link id="108" from="2" to="9" length="1500.00" capacity="40" freespeed="14.88" permlanes="1" />
<link id="109" from="2" to="10" length="1500.00" capacity="40" freespeed="14.88" permlanes="1" />
<link id="110" from="2" to="11" length="1500.00" capacity="40" freespeed="14.88" permlanes="1" />
<link id="111" from="3" to="12" length="750.00" capacity="40" freespeed="14.88" permlanes="1" />
<link id="112" from="4" to="12" length="750.00" capacity="40" freespeed="14.88" permlanes="1" />
<link id="113" from="5" to="12" length="750.00" capacity="40" freespeed="14.88" permlanes="1" />
<link id="114" from="6" to="12" length="750.00" capacity="40" freespeed="14.88" permlanes="1" />
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<link id="215" from="12" to="7" length="750.00" capacity="40" freespeed="14.88" permlanes="1" />
<link id="216" from="12" to="8" length="750.00" capacity="40" freespeed="14.88" permlanes="1" />
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<link id="219" from="12" to="11" length="750.00" capacity="40" freespeed="14.88" permlanes="1" />
<link id="220" from="13" to="12" length="1500.00" capacity="80" freespeed="14.88" permlanes="1" />
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<link id="222" from="15" to="14" length="4500.00" capacity="80" freespeed="14.88" permlanes="1" />
<link id="223" from="1" to="15" length="1500.00" capacity="80" freespeed="14.88" permlanes="1" />
</links>
```

Figures 08.

(i) Presenting the links in MATSim as two-way roads

All the road links in the base scenario are configured as one-way roads. However, majority of the roads in Hong Kong are bidirectional and the number of agents has highly increased in the modified scenario, so, we made a reverse link for each original link in the “equil” scenario and therefore the road links are also changed to two-way roads to increase their capacity. The total number of road links is then increased from 23 to 46 in all modified scenarios.

The direction of the road links is changed by creating new links but with an opposite direction in the network file. The new links will connect to the same road nodes, but a different link id will be assigned to them. Bidirectional road network can better represent the complexity and density of Hong Kong’s transport infrastructure.

(ii) Modifying Capacity

The capacity of each road link is defined in the network file related to a MATSim simulation which was adjusted to reflect different road types and traffic conditions, considering practical assumptions and data references. In Hong Kong, the roads may have capacity of about 2000 vehicles per hour (Transport Department, 2023). The value in capacity represents the maximum number of vehicles that can pass through a particular point of the link per hour. It allows us to test how varying capacities influence congestion and travel behavior.

In scenario 1, as shown in Figure 6 above, capacities were set to 250 and 2500 vehicles/hour for different road lengths. According to the statistics report of the Transport Department in 2023, the ownership of private cars per 1000 population is 77 (Transport Department, 2023). Since the agent number we set is 500 which means there will be around 39 private cars, the capacity of 250 should be more than enough for the simulation network. Actually, Scenario 1 is designed to test road conditions suitable for larger populations. For example, if the number of agents scales up to 2000, this high-capacity road network would handle the increased traffic

without significant congestion. Although the current agent count is low, this setup provides insights into network performance under ideal conditions with ample road capacity.

In scenario 3, as shown in Figure 7 above, the road capacities were reduced to 80 and 40 vehicles/hour for different road lengths, mimicking conditions closest to real-world urban congestion. This scenario aims to mimic real-world road conditions more accurately, drawing from data mentioned in the above, where approximately 77 private cars exist per 1,000 people. For the 500 agents, this translates to an estimated capacity of 40 vehicles/hour per road which reflects highly congested roads, aligning closely with the realities of Hong Kong's dense urban environment.

For scenario 2, as shown in Figure 8 above, the road capacities were increased to 200 and 100 vehicles/hour for different road lengths. This scenario explores the impact of slightly increased capacities compared to Scenario 3. It helps to examine whether moderate improvements in road infrastructure can reduce congestion and improve traffic flow.

(iii) Modifying Length

The length of a link represents the length of a road in meters. It can be used to calculate speed and travel time. The length of a link represents the length of a road in meters. It can be used to calculate speed and travel time. It can also explore how travel distance impacts congestion and agent route selection. The length of public roads per 1000 population is 0.298 km (298 m) (Transport Department, 2023). Shorter roads represent highly dense areas like those commonly found in Hong Kong, providing insights into the behavior of agents in these compact urban environments. Thereby, the lengths of roads in the base "equil" scenario of 10000 m, 5000 m and 35000 m are changed to 1500 m, 750 m respectively.

(iv) Rationale behind the length and capacity modifications – visualize the differences

The rationale behind the modification of the lengths and capacities of the roads, however, unlike the activities of the agents mentioned later in the report, where we adjust and make it simulate real world data, these modifications are made to visualize the differences of the scores under different configurations. For example, the roads in practice might have different lengths and capacities, and it could be hard to generalize them. Despite this, we still try to scale down the capacity and the lengths to make the scenario deviate less from the real world as mentioned above in modifying capacity and length. In conclusion, we take a slightly different approach here and make them into different scenarios.

The freespeed in network.xml represents the speed that vehicles can travel when no congestion happens in a link. The maximum travel speed on all roads in Hong Kong is 50km/h, which is about 13.88 m/s. Hence, the freespeed of the links is changed to 14.88 m/s, 1m/s above the speed limit, to test how changing the freespeed would affect the travel time, congestion, network efficiency and travel behaviors of agents in all modified scenario.

5.1.2 Modification of plan data and reasons

(i) Creating 400 additional agents and their daily plans

We use a total sample of 500 agents using a Python script in all modified scenario, which is the maximum number of agents that can be visualized using the free license of the Via visualization tool, 400 more than the default MATSim plans100.xml activity-based model.

In addition, the 400 newly added agents were assigned with new agent id, but their original plans remained unchanged. Therefore, for the initial iteration of the MATSim simulation, there will be agents with the same initial plans for the day, with travel exclusively by car. This uniformity helps provide a controlled baseline for evaluating subsequent changes and adaptations in agent behaviour throughout the simulation process.

(ii) Activity generation

Each agent in the Python script defines the `home_xy` (starting point), `dest_xy` (destination point), `intent` (activity during the day), `intent_overnight` (activity during the night), `start_time`, and `end_time`. The randomization steps are as follows:

- Starting point: we randomly choose a link and randomly interpolate a point on the link. Then we randomize a point within a certain distance to point on the link.
- Destination point: we randomly choose a facility, then we randomize a point within a certain distance to that facility.
- Day activity/ night activity: we randomly assign a travel purpose to the agent, in respect of the distribution of purposes of our model, night activity is home by default unless the day activity overflows 00:00, then day and night activities are swapped.
- Start time/ end time: we randomly generate a timestamp, with respect to the distribution of our model.

(iii) Rationale behind the modification - data integrity

We aim to simulate a minimalized version of the travel data from the Hong Kong Travel Characteristics Survey in 2011 (the Survey) by Transport Department, therefore the accuracy to represent real world data is one of our major concerns. The following figures shows a comparison of our 500-agent model against the hourly trip-making time from the Survey (Transport Department,2011).

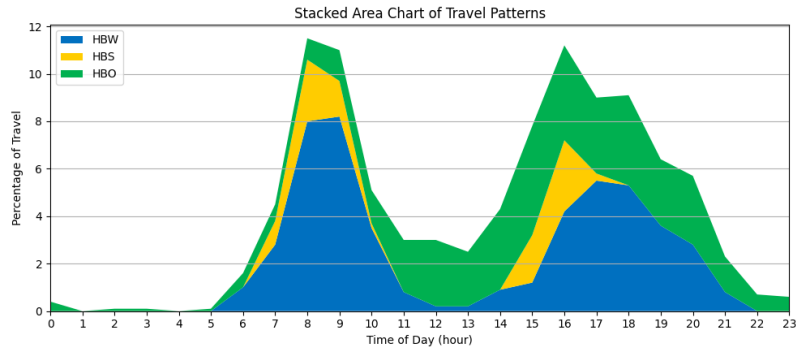


Figure 09. The trip making time of our 500-agent activity-based model.

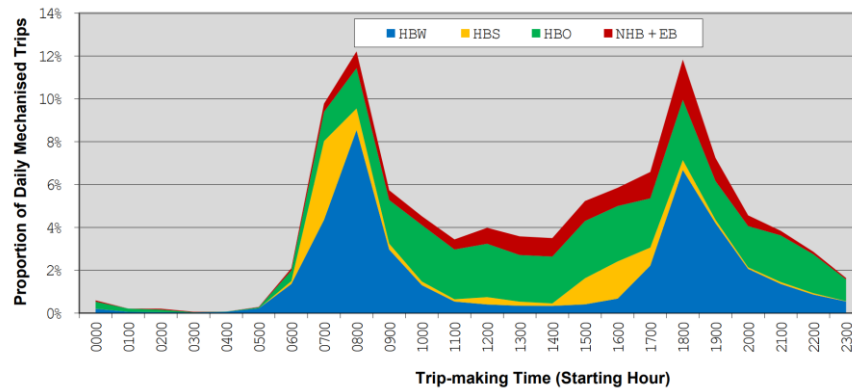


Figure 10. The trip-making time of the agents from the 2011 survey by the Transport Department. (Transport Department, 2011)

The above graphs appear to have some overlaps, and our graph appears to have a section less than the one from the Survey. We believe that our model, to a certain extent, can represent the data from the Transport Department. During our development of the MATSim scenario, we find that developing NHB (Non-home-based) + EB (Employer's business) travel activities, accounting for 13% and 5% of the total trips respectively, are more difficult than expected. They added an extra layer of complexity to the activity schedule, rather than the straightforward home-activity-home pattern. For instance, non-home-based travel activities could involve travel from work or school to other destinations, while Employer's business travel activities represent work-to-work trips. To maintain a simpler MATSim scenario, we have decided to exclude these more complex activity types.

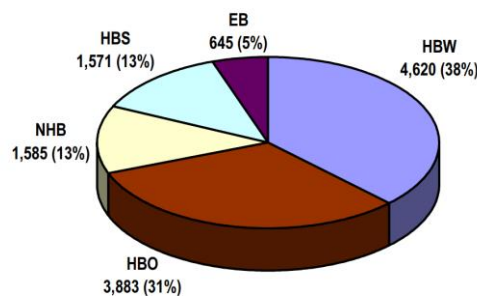


Figure 11. Daily Mechanized trips by purpose (Transport Department, 2011)

5.1.3 Modification of facilities data and reasons

The modifications to facilities data and activity parameters were made to better represent the urban environment and travel behavior in Hong Kong. We believe that facilities should function like districts in Hong Kong, and districts can hold various types of activities, such as residential, commercial, educational, and recreational purposes. Therefore, all facilities in our activity model are set to accept all kinds of activity purposes, including home (h), work (w), school (s), and other activities (o).

5.2 Modified configuration

5.2.1 Modification of iteration number and reasons

The iteration number on config.xml is changed to 20 and 50 to find out the changes of scoring along iterations. The iterations are more than the “equil” scenario to yield a more accurate route planning result due to the randomness caused by the introduced changes. The scoring results may be used to indicate if more iterations are required while the network capacity decreases. Since MATSim is based on an evolutionary algorithm, it depends on randomly generated plans, random choices and scoring of selected and non-selected plans. When we increase the number of agents, it might cause the need of more random replications until an optimum solution is found (Llorca & Moeckel, 2019). Also, in the situation where most of the different route's capacity reduced to 100 vehicles, it may require more iterations for 500 agents to be assigned to the routes.

5.2.2 Modification of activity parameters and reasons

The type of activity including home, work, school and other activities are added with a priority level of 1. The typical duration for home, work and school is set to 8 hours, and that for other activities is set to 15 hours. There is no latestStartTime and minimalDuration set for each activity, allowing the simulation to capture diverse and flexible travel patterns. The table below summarizes the configuration and network parameters in different scenarios briefly.

	Length	Capacity (vehicle/hour)	Free speed (m/s)	Iteration	Agent	Activities
Equil	35000m, 2000m, 10000m, 5000m	36000, 3600, 1000, 360	27.78	10	100	home, work
Scenario 1	10000m, 5000m	2500, 250	14.88	20, 50	500	home, work, school, others
Scenario 2	1500m, 750m	200, 100	14.88	20, 50	500	home, work, school, others
Scenario 3	1500m, 750m	80, 40	14.88	20, 50	500	home, work, school, Others

Table 01. Summary of scenarios

6. Results and Comparison

The following shows the customized MATSim scenarios compared to the original “equil” scenario to understand how the modifications influence the simulation outcomes:

6.1 Result and Comparison of Travel Speeds

Scenario	Freespeed	Average travel distance	Average trip duration	Average travel speed
Baseline (equil)	27.78	44776.119	27:08	27.503
“equil”, freespeed changed to 14.88	14.88	44776.119	50:28	14.787
“equil”, with freespeed 14.88, 500 customized agents	14.88	65160	01:13:03	14.867
Scenario 1	14.88	19975	22:25	14.851
Scenario 2	14.88	3022.523	03:26	14.672
Scenario 3	14.88	3144	03:44	14.036

Table 02. Comparison of scenarios

In the baseline "equil" scenario, the travel speed was 27.503 m/s, reflecting the idealized conditions of a simple network with a freespeed of 27.78 m/s. When the freespeed was reduced to 14.88 m/s, the travel speed in the baseline dropped to 14.787 m/s. This shows the direct impact of a lower speed limit on agent mobility.

We have also introduced 500 customized agents to the baseline network with a freespeed of 14.88 m/s, resulting in a slightly higher travel speed of 14.867 m/s. The slightly increase in

travel speed indicates that the increased number of agents actually had a limited effect on congestion in the simplified network.

In the customized scenarios, travel speeds varied further due to modifications in road lengths and capacities. In Scenario 1, it featured the longest and highest-capacity roads (250 and 2500 vehicles/hour). Therefore, the travel speed was 14.851 m/s, demonstrating smooth traffic flow under minimal congestion. In Scenario 2, the road lengths are shorter and the capacities (200 and 100 vehicles/hour) are reduced. Thus, the travel speed is decrease to 14.672 m/s, reflecting increased congestion due to the more constrained network. Scenario 3, designed to mimic real-world urban congestion with the shortest roads and lowest capacities (80 and 40 vehicles/hour). It showed the lowest travel speed of 14.036 m/s, emphasizing the significant impact of limited road infrastructure on traffic efficiency.

To conclude, from all the comparisons of the results, we found out that travel length affects the average travel distance, and travel speed is dependent on distance, capacity, and freespeed. Freespeed is the biggest factor in travel speed, followed by capacity.

6.2 Visualization

We can visualize our MATSim scenario by adding `network.xml`, `facilities.xml`, and `output_events.xml.gz` to Via, then click Load Data.

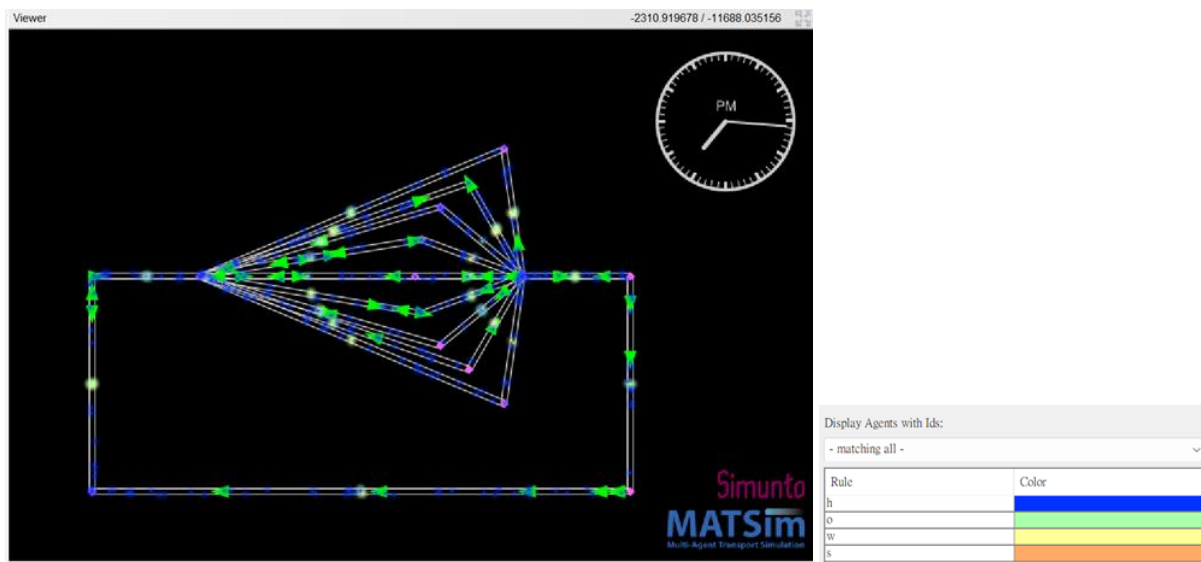


Figure . A screenshot of our MATSim scenario in Via at around 07:15, a peak hour for work and school activities

6.3 Comparison of effects of iteration and average travel speed

According to Table 3, it can be observed that the travel distance and the travel time remain relatively constant across iterations 0 to 50. This might be contributed to the network configuration and pattern of agent behavior.

Iteration	Average travel distance	Average travel time
0	3144	03:46
10	3144	03:45
20	3144	03:44
30	3144	03:44
40	3144	03:44
50	3144	03:44

Table 03. Comparison of average travel time with different iterations in scenario 3

In our modified scenarios, the network structure is relatively simple with limited route choices. The reduced road lengths (from 35000m to 1500m in Scenarios 2 and 3) and adjusted capacities create a constrained environment where alternative routes may not offer significant advantages. Due to the compact network design representing Hong Kong's dense urban environment, agents have limited opportunities to find substantially different routes.

Furthermore, the fixed activity locations and time windows for activities (8 hours for work/school) of agent behavior may limit the potential optimization through alternative departure times or routes. With 500 agents sharing limited road capacity, the system quickly reaches a stable state where further iterations do not produce meaningful improvements.

The balance between network capacity and number of agents creates a relatively stable traffic flow pattern from early iterations. This indicates that our simplified Hong Kong scenario may be sufficient to achieve stable results in a lower number of iterations (around 20).

6.4 Result and Comparison of Score of Daily Plan between 20 iterations and 50 iterations

Scenario	Score of 20 iterations (average executed)	Score of 50 iterations
Equil	67.1664	67.4053
Scenario 1	103.3626	104.0953
Scenario 2	110.0388	110.0386
Scenario 3	109.9846	109.9852

Table 04. Comparison of Score of Daily Plan between 20 and 50 iterations

Finding: The scores of iterations 20 and 50 are similar with small difference

We conclude that the scoring depends little on the iterations but greatly depends on the freespeed and road capacity. It may contribute to the network configuration and pattern of agent behavior mentioned above in 6.3, and the number sample of agents is relatively small compared to real world situation. Hence, about 20 iterations are effective for optimizing the score of daily plans.

7. Conclusion

7.1 Key Findings

Our customized MATSim scenarios have provided several insights into the impacts of road network modifications, agent configurations, and travel parameters on urban transportation dynamics in Hong Kong. Detail findings will be explored in follow:

7.1.1 Network Capacity Impact

The modifications of road capacity (from 36000 to 250/200/40 vehicles/hour) significantly affected simulation outcomes. Lower capacities resulted in increased congestion and longer travel times, while higher capacities showed improved traffic flow. However, the low-capacity network (Scenario 3) is considered the better reflect to the realistic Hong Kong's urban conditions compared with other scenarios, since its capacity (40 vehicles/hour) creates realistic congestion patterns and aligns with Hong Kong's actual urban density challenges.

7.1.2 Iteration and Optimization

The scores of daily plans across 20 and 50 iterations remained relatively stable which means that 20 iterations were sufficient for achieving optimized plans in this simulation. This stability highlights the limited benefits of further iterations in compact networks with constrained route choices, as seen in Hong Kong's urban context.

7.1.3 Agent Activity Patterns

The agent activity patterns play an important role in creating realistic urban mobility scenarios and understanding traffic flow dynamics. Through diverse agent activity patterns, it improves the accuracy of urban transportation representations, understanding of peak hour dynamics, and evaluating infrastructure needs.

7.1.4 Effect of Freespeed

Freespeed proved to be the most critical factor influencing travel speed. Reducing the freespeed in the baseline "equil" scenario from 27.78 m/s to 14.88 m/s resulted in a significant drop in travel speed from 27.503 m/s to 14.787 m/s. Across all customized scenarios, the uniform freespeed of 14.88 m/s led to consistent travel patterns, emphasizing its dominant role in agent mobility.

7.2 Limitations

7.2.1 Simplicity of network

The relatively small sample of agents, the simplicity of the "equil" network, and the uncertainty in the travel behavior assumptions are one of the limitations. Also, the 500-agent activity model excluded the more complex NHB and EB travel activities, which account for a notable 13% and 5% of total trips respectively. This could further limit the model's ability to accurately represent the true travel patterns. 500 agents are a miniscule representation of around 12 million trips on an average weekday recorded by the Transport Department (2011).

While the model may not perfectly match the Transport Department's graphs in every detail, it is able to moderately capture the key trends and patterns. For example, our model is able to capture the peak hours for home-based work, school, and other, both the start and end. The decision to exclude the more complex NHB and EB activities was made strategically to maintain a simpler, more manageable MATSim scenario, which can provide valuable insights and serve as a solid foundation for further model refinement and development.

Considering the tradeoffs and the model's ability to reasonably represent the broader transportation data, we conclude that our 500-agent activity-based model is a good enough representation to proceed with the current MATSim scenario development. The benefits of a simpler, more focused model outweigh the limitations imposed by the exclusion of the NHB and EB activities, and we believe this approach will ultimately lead to a more robust and practical transportation planning tool.

7.2.2 Modifying Parameters in Scenarios

Another limitation is the length and the capacity of each road, in this project instead of making adjustments to match the real-world data, we find that the road conditions are hard to generalize, so we have resort to a different approach - creating 3 scenarios, each with varying length and capacity of the roads. We did try to scale down the lengths and capacities to match with the 500-agent model, but without reference to any data, the accuracy is far from realistic. We do have some numbers from the Transport Department for example the total public transport patronage in December 2023 was around 328 million, and the total public road length was around 2200 km (Transportation Department, 2023), theoretically we could calculate the demand per road length, but the supply, or in other words the vehicle capacity per kilometer, is still unknown.

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